## 17.10 THE APPLICATION OF QUEUEING THEORY

Because of the wealth of information provided by queueing theory, it is widely used to guide the design (or redesign) of queueing systems. We now turn our focus to how queueing theory is applied in this way.

The most common decision that needs to be made when designing a queueing system is how many servers to provide. However, a number of other decisions also may be needed. The possible decisions include

1. Number of servers at a service facility.
2. Efficiency of the servers.
3. Number of service facilities.

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4. Amount of waiting space in the queue.
5. Any priorities for different categories of customers.

The two primary considerations in making these kinds of decisions typically are (1) the cost of the service capacity provided by the queueing system and (2) the consequences of making the customers wait in the queueing system. Providing too much service capacity causes excessive costs. Providing too little causes excessive waiting. Therefore, the goal is to find an appropriate trade-off between the service cost and the amount of waiting.

Two basic approaches are available for seeking this trade-off. One is to establish one or more criteria for a satisfactory level of service in terms of how much waiting would be acceptable. For example, one possible criterion might be that the expected waiting time in the system should not exceed a certain number of minutes. Another might be that at least 95 percent of the customers should wait no longer than a certain number of minutes in the system. Similar criteria in terms of the expected number of customers in the system (or the probability distribution of this number) also could be used. The criteria also might be stated in terms of the waiting time or the number of customers in the *queue* instead of in the system. Once the criterion or criteria have been selected, it then is usually straightforward to use trial and error to find the least costly design of the queueing system that satisfies all the criteria.

The other basic approach for seeking the best trade-off involves assessing the costs associated with the consequences of making customers wait. For example, suppose that the queueing system is an *internal service system* ( as described in Sec. 17.3), where the customers are the employees of a for-profit company. Making these employees wait at the queueing system causes *lost productivity,* which results in *lost profit.* This lost profit is the **waiting cost** associated with the queueing system. By expressing this waiting cost as a function of the amount of waiting, the problem of determining the best design of the queueing system can now be posed as minimizing the expected *total cost* (service cost plus waiting cost) per unit time.

We spell out this latter approach below for the problem of determining the optimal number of servers to provide.

### How Many Servers Should Be Provided?

To formulate the objective function when the decision variable is the number of servers $s$, let

$E(\text{TC})$ = expected total cost per unit time,

$E(\text{SC})$ = expected service cost per unit time,

$E(\text{WC})$ = expected waiting cost per unit time.

Then the objective is to choose the number of servers so as to

$$\text{Minimize } E(\text{TC}) = E(\text{SC}) + E(\text{WC}).$$

When each server costs the same, the **service cost** is

$$E(\text{SC}) = C_s s,$$

where $C_s$ is the marginal cost of a server per unit time. To evaluate WC for any value of $s$, note that $L = \lambda W$ gives the expected total amount of waiting in the queueing system per unit time. Therefore, when the waiting cost is proportional to the amount of waiting, this cost can be expressed as

$$E(\text{WC}) = C_w L,$$

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where $C_w$ is the waiting cost per unit time for each customer in the queueing system. Therefore, after estimating the constants, $C_s$ and $C_w$, the goal is to choose the value of $s$ so as to

$$\text{Minimize } E(\text{TC}) = C_s s + C_w L.$$

By choosing the queueing model that fits the queueing system, the value of $L$ can be obtained for various values of $s$. Increasing $s$ decreases $L$, at first rapidly and then gradually more slowly.

Figure 17.13 shows the general shape of the $E(\text{SC})$, $E(\text{WC})$, and $E(\text{TC})$ curves versus the number of servers $s$. (For better conceptualization, we have drawn these as smooth curves even though the only feasible values of $s$ are $s = 1, 2, \ldots$.) By calculating $E(\text{TC})$ for consecutive values of $s$ until $E(\text{TC})$ stops decreasing and starts increasing instead, it is straightforward to find the number of servers that minimizes total cost. The following example illustrates this process.

### An Example

The Acme Machine Shop has a tool crib to store tools required by the shop mechanics. Two clerks run the tool crib. The clerks hand out the tools as the mechanics arrive and request them. The tools then are returned to the clerks when they are no longer needed. There have been complaints from supervisors that their mechanics have had to waste too much time waiting to be served at the tool crib, so it appears as if there should be *more* clerks. On the other hand, management is exerting pressure to reduce overhead in the plant, and this reduction would lead to *fewer* clerks. To resolve these conflicting pressures, an OR study is being conducted to determine just how many clerks the tool crib should have.

The tool crib constitutes a queueing system, with the clerks as its servers and the mechanics as its customers. After gathering some data on interarrival times and service times, the OR team has concluded that the queueing model that fits this queueing system best is the *M/M/s* model. The estimates of the mean arrival rate $\lambda$ and the mean service rate (per server) $\mu$ are

$$\lambda = 120 \text{ customers per hour,}$$
$$\mu = 80 \text{ customers per hour.}$$

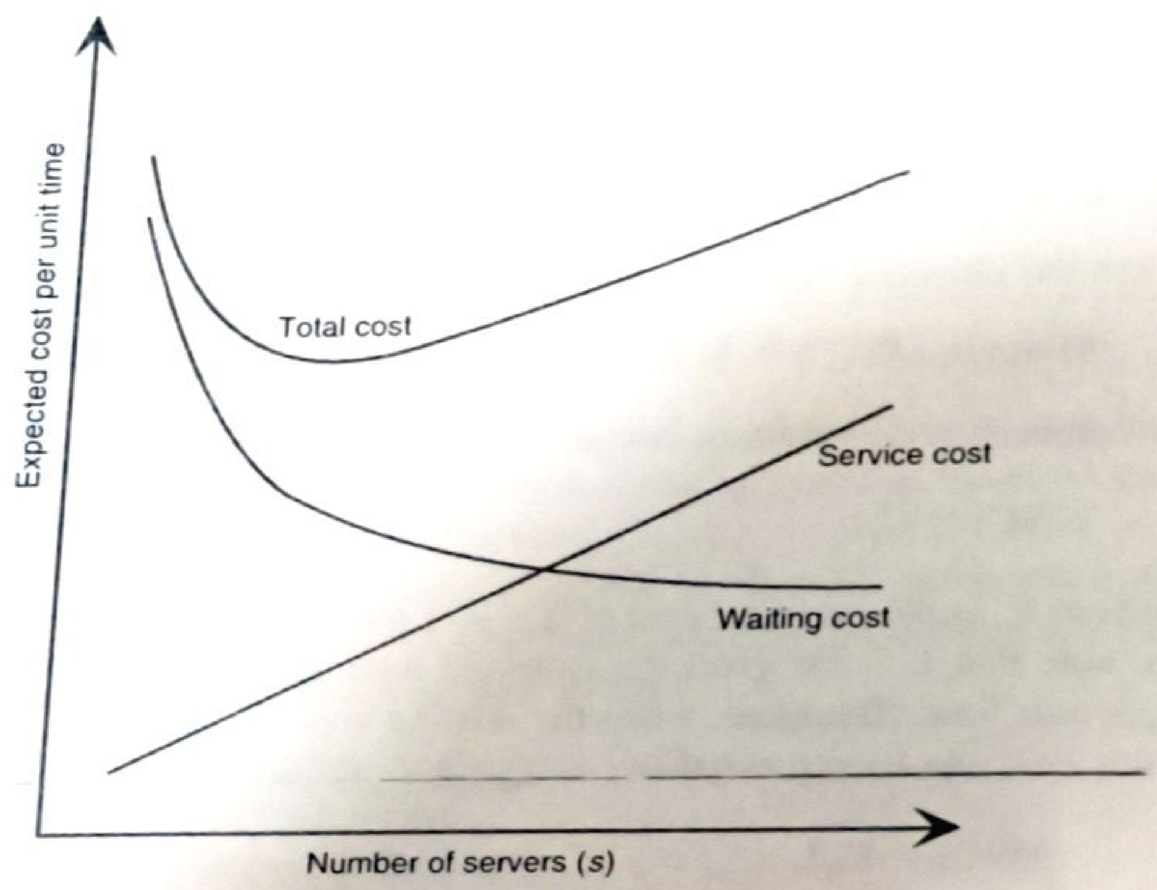


**FIGURE 17.13**
The shape of the expected cost curves for determining the number of servers to provide.

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so the utilization factor for the two clerks is

$$\rho = \frac{\lambda}{s\mu} = \frac{120}{2(80)} = 0.75.$$

The total cost to the company of each tool crib clerk is about \$20 per hour, so $C_s$ = \$20. While a mechanic is busy, the value to the company of his or her output averages about \$48 per hour, so $C_w$ = \$48. Therefore, the OR team now needs to find the number of servers (tool crib clerks) $s$ that will

Minimize $E(\text{TC}) = \$20\, s + \$48\, L$.

An Excel template has been provided in your OR Courseware for calculating these costs with the *M/M/s* model. All you need to do is enter the data for the model along with the unit service cost $C_s$, the unit waiting cost $C_w$, and the number of servers $s$ you want to try. The template then calculates $E(\text{SC})$, $E(\text{WC})$, and $E(\text{TC})$. This is illustrated in Fig. 17.14 with $s = 3$ for this example. By repeatedly entering alternative values of $s$, the template then can reveal which value minimizes $E(\text{TC})$ in a matter of seconds.

Table 17.5 shows the data that would be generated from this template by repeating these calculations for $s = 1, 2, 3, 4$, and 5. Since the utilization factor for $s = 1$ is $\rho = 1.5$, a single clerk would be unable to keep up with the customers, so this option is ruled out. All larger values of $s$ are feasible, but $s = 3$ has the smallest expected total cost. Furthermore, $s = 3$ would decrease the current expected total cost for $s = 2$ by \$61 per hour. Therefore, despite management's current drive to reduce overhead (which includes the cost of tool crib clerks), the OR team recommends that a third clerk be added to the tool crib. Note that this recommendation would decrease the utilization factor for the clerks from an already modest 0.75 all the way down to 0.5. However, because of the large improvement in the productivity of the mechanics (who are much more expensive than the clerks) through decreasing their time wasted waiting at the tool crib, management adopts the recommendation.

## Other Issues

Chapter 26 on the book's website expands considerably further on the application of queueing theory, including how to deal with some other issues not considered above.

For example, the analysis displayed in Fig. 17.14 and Table 17.5 assumed that the waiting cost is proportional to the amount of waiting, but this sometimes is not the case. If a company has one or two of its employees in a queueing system, this may not be very serious in terms of their lost productivity because others may be able to handle all of the available productive work. However, having additional employees in the queueing system may result in a sharp increase in lost productivity and the resulting lost profit, so the waiting cost becomes a nonlinear function of the number in the system. Similarly, the consequences to a commercial service system for making its customers wait may be minimal for short waits but much more serious for long waits. In this case, the waiting cost becomes a nonlinear function of the waiting time. Section 26.3 describes the formulation of nonlinear waiting-cost functions and then the calculation of $E(\text{WC})$ with such functions.

Section 26.4 discusses a decision model where the decision variables are *both* the number of servers and the mean service rate for the servers. An interesting issue that arises here is whether it is better have *one fast server* (several people working together to serve each customer rapidly) or *several slow servers* (several people working separately to serve different customers).

Section 26.4 also presents a decision model where the decision variables are the number of service facilities and the number of servers per facility to provide service to a calling population of potential customers. Given the mean arrival rate for the entire calling population, increasing the number of facilities enables decreasing the mean arrival rate

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| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| 1 | Economic Analysis of Acme Machine Shop Example | | | | | | |
| 2 | | | | | | | Results |
| 3 | | | Data | | | L = | 1.736842105 |
| 4 | | λ = | 120 | (mean arrival rate) | | $L_q$ = | 0.236842105 |
| 5 | | μ = | 80 | (mean service rate) | | | |
| 6 | | s = | 3 | (# servers) | | W = | 0.014473684 |
| 7 | | | | | | $W_q$ = | 0.001973684 |
| 8 | | Pr(W > t) = | 0.02581732 | | | | |
| 9 | | when t = | 0.05 | | | ρ = | 0.5 |
| 10 | | | | | | | |
| 11 | | Prob($W_q$ > t) = | 0.00058707 | | | n | $P_n$ |
| 12 | | when t = | 0.05 | | | 0 | 0.210526316 |
| 13 | | | | | | 1 | 0.315789474 |
| 14 | | Economic Analysis: | | | | 2 | 0.236842105 |
| 15 | | Cs = | $20.00 | (cost / server / unit time) | | 3 | 0.118421053 |
| 16 | | Cw = | $48.00 | (waiting cost / unit time) | | 4 | 0.059210526 |
| 17 | | | | | | 5 | 0.029605263 |
| 18 | | Cost of Service | $60.00 | | | 6 | 0.014802632 |
| 19 | | Cost of Waiting | $83.37 | | | 7 | 0.007401316 |
| 20 | | Total Cost | $143.37 | | | | |

| | B | C |
|---|---|---|
| 18 | Cost of Service | =Cs*s |
| 19 | Cost of Waiting | =Cw*L |
| 20 | Total Cost | =CostOfService+CostOfWaiting |

| Range Name | Cells |
|---|---|
| CostOfService | C18 |
| CostOfWaiting | C19 |
| Cs | C15 |
| Cw | C16 |
| L | G4 |
| s | C6 |
| TotalCost | C20 |

**FIGURE 17.14**
This Excel template for using economic analysis to choose the number of servers with the *M/M/s* model is applied here to the Acme Machine Shop example with $s = 3$.

**TABLE 17.5** Calculation of $E(TC)$ for alternative $s$ in the Acme Machine Shop example

| $s$ | $\rho$ | $L$ | $E(SC) = C_s s$ | $E(WC) = C_w L$ | $E(TC) = E(SC) + E(WC)$ |
|---|---|---|---|---|---|
| 1 | 1.50 | ∞ | $20 | ∞ | ∞ |
| 2 | 0.75 | 3.43 | $40 | $164.57 | $204.57 |
| 3 | 0.50 | 1.74 | $60 | $83.37 | $143.37 |
| 4 | 0.375 | 1.54 | $80 | $74.15 | $154.15 |
| 5 | 0.30 | 1.51 | $100 | $72.41 | $172.41 |

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